

Lower Bounds on Lattice Triangulations

A Simulation Study of Lower Bounds on the Mixing Time of Glauber
Dynamics on Lattice Triangulations

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1 Introduction

Lattice triangulations are fundamental geometric objects equipped with beautiful structures; see [3], for background. A lot of its structural properties have been studied extensively in earlier papers. For instance, several recent papers have developed ingenious combinatorial arguments to estimate the asymptotic number of triangulations [1, 5, 7, 8], as well as to explore their connectivity properties under local moves [4, 5]. Interests in lattice triangulations also arise from the algebraic geometry community through connections with plane algebraic curves and Hilbert's 16th problem [9].

A natural generalization of the model is to introduce weights. In particular, we consider the distribution in which a triangulation σ is associated with a weight $\lambda^{|\sigma|}$ where $|\sigma|$ is the total l_1 -length of the edges in σ and $\lambda > 0$ is a real parameter. This generalization of the model corresponds rather closely to a model that has been studied statistical physics. An interesting feature of the model is that for different values of λ , edges in a triangulation seem to exhibit different behaviors. See Figure 1.

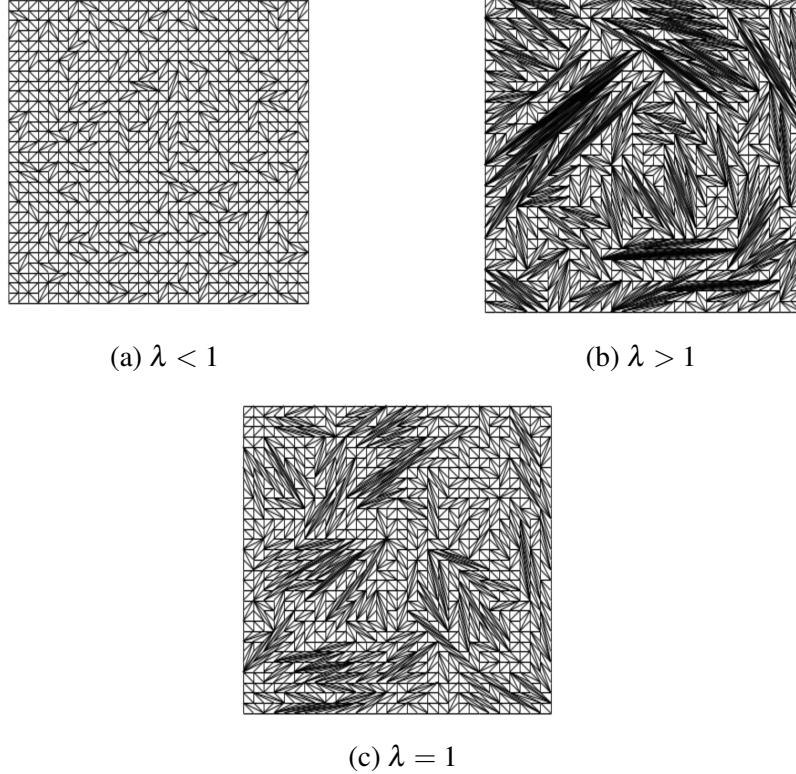


Figure 1: Random triangulations of a 30×30 lattice.

As we can see from the figures above, when $\lambda > 1$, long edges are favored and these edges tend to form large regions of similar slope; when $\lambda < 1$ these regions disappear and

short edges are favored. The case where $\lambda = 1$ seems to represent a "phase transition" between these two regimes where long edges are beginning to form but are not stable.

To generate random triangulations and to quantitatively study the behaviors of random triangulations, we simulate a local Markov Chain, or *Glauber dynamics*, which converges to the desired probability distribution over the triangulations. The Glauber dynamics works as follows: pick a edge uniformly at random, and if it is the diagonal of a convex quadrilateral (which is in fact always a parallelogram), flip it to the opposite diagonal edge. This induces a so-called *flip graph* on the set of triangulations in which two triangulations share an edge if and only if one can be obtained from the other by one edge flip.

This leads to a fundamental question of both algorithmic and structural interest, posed by Welzl and others [10]: what is the *mixing time* of the Glauber dynamics? The distinct behaviors we observed in Figure 1 as a result of the different values of λ can also be reflected through the lens of the mixing time of the Glauber dynamics. Specifically, Caputo, Martinelli, Sinclair and Stauffer in their paper [2] posed the following conjecture:

Conjecture 1.1. (a) For any fixed $\lambda < 1$, the mixing time of the Glauber dynamics is poly($m + n$) [specifically, $O(mn(m + n))$]. (b) For any fixed $\lambda > 1$, the mixing time is exponential in $(m + n)$ [specifically, $\exp(\Omega(mn(m + n)))$].

In the same paper [2], they proved the $O(mn(m + n))$ upper bound on the mixing time of the Glauber dynamics for $\lambda \leq 1/8$. They also gave a proof of the current best-known $\exp(\Omega((m + n)))$ lower bound on the mixing time of the Glauber dynamics for $\lambda > 1$ which we will reproduce in Section 3. However, notice that there is a gap between this lower bound and the conjectured lower bound. In this thesis, we will mainly focus on how to obtain a better lower bound for $\lambda > 1$. Specifically, we will go through a series of simulation studies to experimentally identify some key properties of the Glauber dynamics for $\lambda > 1$.

We will study the long-term behaviors of the *herringbone* structures defined in the proof of the current best lower bound in [2]. We will also study the behaviors of the *unidirectional* herringbone structure which is similar to the herringbone structure. By studying and comparing the behaviors of these two structure, we observe that one set of stable triangulations in the case where $\lambda > 1$ is the set of triangulations with long diagonal edges oriented in the same direction. Following this observation, we define a set of triangulations with the so-called *fan structure*. Each triangulation with the fan structure has large average edge length and all edges are oriented in the same direction. The long-term behaviors of the Glauber dynamics on the fan structure suggest that triangulations with long diagonal edges oriented in the same direction are indeed stable. Lastly, we use these observations to propose possible adaptations of the proof of the current best lower bound in [2] to obtain better lower bounds for $\lambda > 1$.

2 The model

In this section, we will introduce the necessary definitions and properties of lattice triangulations that will be used throughout this paper. We follow the same notations as in [2].

2.1 Lattice triangulations

Let $\Lambda_{m,n}^0 := \{0, 1, \dots, m\} \times \{0, 1, \dots, n\}$ denote the set of points in the $m \times n$ rectangular region of the integer lattice $\mathbb{Z}^2$. An edge e in a triangulation is a line segment connecting two vertices $v_1, v_2 \in \Lambda_{m,n}^0$ such that e does not go through any other vertex $u \in \Lambda_{m,n}^0$. A (full) triangulation $\Omega_{m \times n}$ of $\Lambda_{m,n}^0$ is then any maximal set of noncrossing edges. (We will sometimes drop the subscripts on Λ^0 and Ω for simplicity when their values are not important.) We will use the terms triangulation and configuration interchangeably throughout this paper.

Now we describe some basic properties of lattice triangulations. First notice that there are a total of $(m+1)(n+1)$ points, $2mn$ triangles and $3mn + m + n$ edges, $2(m+n)$ of which are on the boundary and the remainder in the interior. Moreover, it follows from Pick's theorem that every triangle has area $\frac{1}{2}$ since $\Lambda_{m,n}^0$ consists of the integer points inside a lattice polygon.

In each triangulation, the *midpoints* of the edges are precisely the half-integer lattice with the integer points removed, that is,

$$\Lambda_{m,n} = \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots, m - \frac{1}{2}, m\} \times \{0, \frac{1}{2}, 1, \dots, n - \frac{1}{2}, n\} \setminus \Lambda_{m,n}^0,$$

namely, the set of half-lattice points in the $m \times n$ rectangle such that at least one of the two entries are half-integers. With this definition, we observe that there is a natural bijection between edges and *midpoints*. To see this, notice that each midpoint is on one and only one edge since the set of noncrossing edges is maximal. On the other hand, each edge can be identified by its midpoint (i.e. the average of its endpoints). This midpoint does not belong to $\Lambda_{m,n}^0$ since otherwise the edge would go through at least three integer points.

Therefore, any triangulation can be fully characterized as an assignment $\sigma = \{\sigma_x\}_{x \in \Lambda_{m,n}}$ of an edge to each point of $\Lambda = \Lambda_{m,n}$. We call midpoints x with one integer and one half-integer coordinate “Type 1,” and those with two half-integer coordinates “Type 2.” Figure 2 shows some examples of valid configurations of edges at Type 1 and 2 midpoints x .

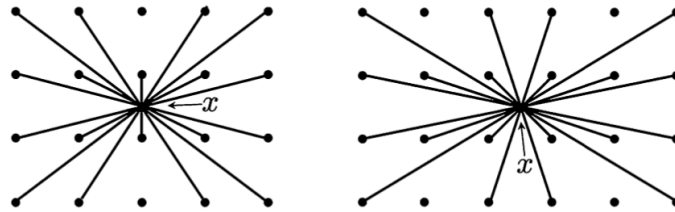


Figure 2: (From [2]) Possible configurations of the edge at x . The configurations are slightly different based on whether x is Type 1 (left) or Type 2 (right).

As we can see from Figure 2, the minimum length configuration of a Type 1 point is vertical or horizontal while that of a Type 2 point is one of the two unit diagonals. The configurations of the boundary edges at points $x = (0, i), (m, i)$ [resp., $x = (j, 0), (j, n)$] are forced to be vertical (resp., horizontal) in all triangulations. We call triangulations in which all edges have minimal length *ground state* triangulations. There are exactly 2^{mn} ground

state triangulations, all of which are equivalent up to flipping of unit diagonals. Figure 3 gives an example of a ground state triangulation.

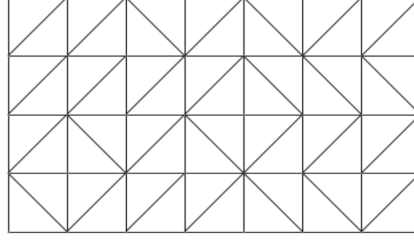


Figure 3: A ground state triangulation of a 4×7 region

We also consider a generalization to lattice triangulations with the configurations of some edges fixed. Let Δ be an arbitrary subset of Λ , and $\eta = \{\eta_x\}_{x \in \Delta}$ an arbitrary assignment of edges to the points of Δ that is consistent; that is, none of the edges in η cross. We denote by $\Omega(\eta, \Delta)$ the set of triangulations $\sigma \in \Omega$ that agree with η on Δ , that is, all possible completions of the partial triangulation η that are consistent with η . Since triangulations are just maximal consistent sets of edges, it is clear that for any η a such completion always exists. We refer to the edges in η as *constraints*. Figure 4 gives an example of triangulations with constraints.

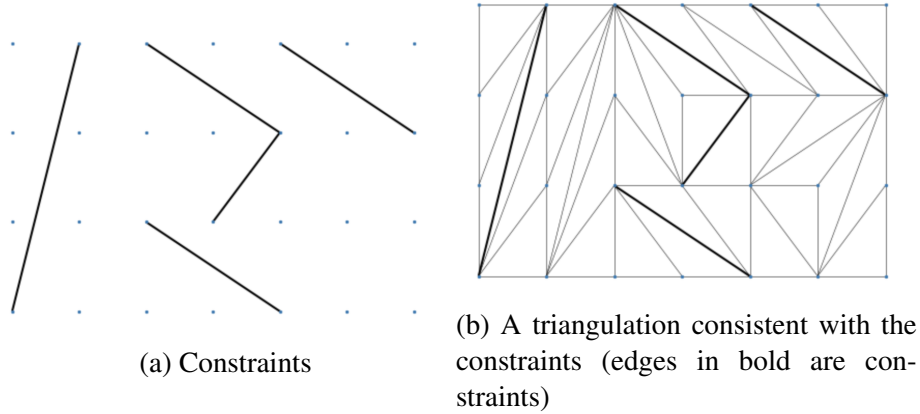


Figure 4: Example of a constrained triangulation of a 3×6 region

2.2 The flip graph

We already know that the boundary edges of lattice triangulations of rectangular regions are fixed. For any interior edge σ_x , σ_x lies in two triangles. The two triangles together form a unique quadrilateral containing σ_x . To satisfy the unimodular property of triangulations, the third vertex of each of these triangles is an integer point on the line parallel to σ_x that passes through the closest integer point on either side of σ_x ; see Figure 5.

The integer points on these lines occur periodically at intervals equal to the length of σ_x and are positioned symmetrically on either side of σ_x . Therefore, given two points a and b

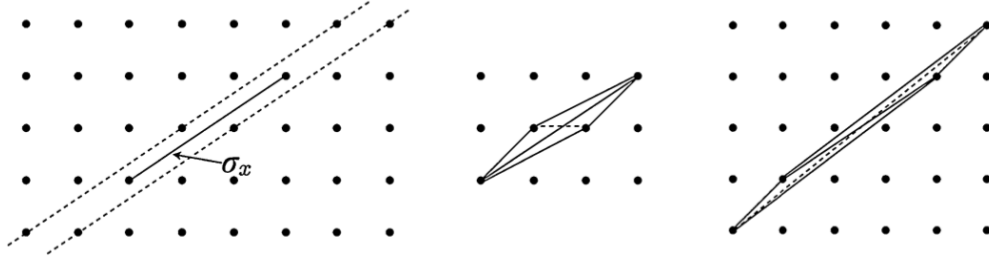


Figure 5: Flips of an edge σ_x . Left: possible locations of the third vertex containing σ_x (the dotted lines); Middle: a flip of σ_x to a shorter edge; Right: a flip of σ_x to a longer edge

on the left and the right closet line to σ respectively, if the line segment $\overline{ab}$ passes through x , then a and b are symmetric about x . In this case, σ_x is a diagonal of the parallelogram and we can replace σ_x with the other diagonal to yield another valid triangulation σ' (σ and σ' differ only at the edge containing x). We say such a σ_x is *flippable*. We observe that, for any σ_x that is not of minimum length, there is a *unique* parallelogram in which σ_x is the longer diagonal; we call this the *minimal parallelogram* of σ_x . Therefore, there exists a unique flip that takes σ_x to a *shorter* edge. (There are many possible flips that make σ_x longer.)

Another observation is that flipping an edge cannot change its orientation from positive to negative unless it is a unit diagonal. In other words, in order to flip the orientation of an edge σ_x , it has go through a sequence of flips that must pass through the minimum configuration of the edge (horizontal, vertical or unit diagonal).

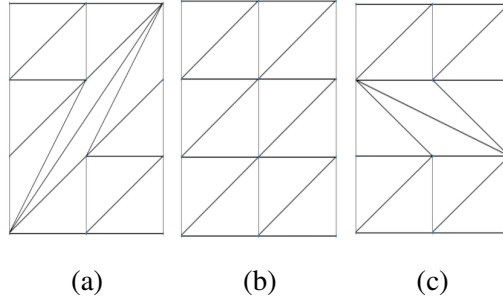


Figure 6: Example of a path that connects two triangulations. Triangulation (a) can be flipped to (c) through a sequence flips that go through (b).

With the flipping operation, we can define the so-called “flip graph” on the set of all triangulations of $\Lambda_{m,n}$. The vertices of the flip graph correspond to triangulations. There exists an undirected edge between two triangulations σ, τ if and only if they differ by one edge flip. The flip graph is connected. To see this, notice that the longest edge in any non-ground-state triangulation is always flippable to a shorter edge since σ_x is the longest edge in both of its triangles, hence the triangles must form the minimal parallelogram of σ_x . As a result, any triangulation σ can reach a ground state after a finite sequence of flips. Since all flips are reversible, we can also reach any triangulation from a ground state. Moreover, any ground state is reachable from any other ground state by flipping unit diagonals. Thus,

for any pair of configurations (σ, τ) , there exists a path from σ to τ corresponding to a sequence of flips from σ to a ground state and then a sequence of flips from that ground state to τ . See Figure 6.

2.3 Weighted triangulations and Glauber Dynamics

To any triangulation σ , we associate a weight $\lambda^{|\sigma|}$ where λ is a real positive parameter and $|\sigma|$ is the total length of σ . We define $|\sigma| = \sum_{x \in \Lambda} |\sigma_x|$ where $|\cdot| : \Lambda \rightarrow \mathbb{Z}^+$ maps an edge to its l_1 -length. These weights induce a *Gibbs distribution* μ defined on Ω via

$$\mu(\sigma) = \frac{\lambda^{|\sigma|}}{Z}$$

where the normalizing factor $Z = \sum_{\sigma \in \Omega} \lambda^{|\sigma|}$ is the *partition function*. The case $\lambda = 1$ corresponds to a uniform distribution on all triangulations; the case $\lambda < 1$ favors shorter edges; the case $\lambda > 1$ favors longer edges. Recall that the ground states we previously defined are triangulations such that all edges are in their minimum length (those that minimize $|\sigma|$). These are precisely the most likely triangulations based on the Gibbs distribution in the scenario when $\lambda < 1$. This explains the choice of the term "ground state." In statistical physics, ground states are configurations of minimum energy, and thus maximum weight in the Gibbs distribution. Note that the term is less appropriate in the case $\lambda > 1$ since in this scenario, triangulations of smaller length are less probable than those of larger length. However, for simplicity, we will use the term "ground state" for triangulations that have minimum total length for the rest of this paper.

With the flip graph and the Gibbs distribution, we can now define a local Markov Chain (or *Glauber dynamics*) on Ω [or, more generally, on $\Omega(\eta, \Delta)$] as follows. In state $\sigma \in \Omega$:

1. Pick a point $x \in \Lambda$ uniformly at random.
2. If σ_x is not flippable, do nothing.
3. Else, if σ_x is flippable to some σ'_x (producing a new triangulation σ'), then flip it with probability

$$\frac{\mu(\sigma')}{\mu(\sigma') + \mu(\sigma)} = \frac{\lambda^{|\sigma'_x|}}{\lambda^{|\sigma'_x|} + \lambda^{|\sigma_x|}}.$$

Since the flip graph is connected, this so-called "heat bath" dynamics defines an ergodic Markov chain on Ω [or on $\Omega(\eta, \Delta)$] that is reversible with respect to μ . Consequently, the dynamics converges to the stationary distribution μ . An important question now is how fast will the dynamics converge to the stationary distribution. We will analyze this rate of convergence by using the standard notion of *mixing time*, defined by

$$T_{mix} = \inf \left\{ t \in \mathbb{N} : \max_{\sigma \in \Omega} \|p^t(\sigma, \cdot) - \mu\| < \frac{1}{4} \right\},$$

where $p^t(\sigma, \cdot)$ denotes the distribution after t steps when the initial state is σ , and

$$\|v - u\| := \frac{1}{2} \sum_{\sigma \in \Omega} |v(\sigma) - u(\sigma)| = \max_{A \subset \Omega} |v(A) - u(A)|$$

is the usual total variation distance between two distributions μ, v .

2.4 Upper bounds on mixing time for $\lambda < 1$

While we will focus mainly on lower bounds on the mixing time for $\lambda > 1$, we briefly state an upper bound result on mixing time for $\lambda < 1$ from [2].

Theorem 2.1. *There exist constants $C > 0$ and $\lambda_0 \in (0, 1)$ such that for all $\lambda \leq \lambda_0$, all $m, n \in \mathbb{N}$,*

$$T_{\text{mix}} \leq Cmn(m+n)$$

In their proof, Caputo, Martinelli, Sinclair and Stauffer defined a distance function between adjacent pairs of triangulations σ, τ for some fixed positive number $\alpha > 1$. If σ_x, τ_x are opposite unit diagonals, define $\Delta(\sigma, \tau) = \alpha^2 - 1$; otherwise let

$$\Delta(\sigma, \tau) = |\alpha^{|\sigma_x|} - \alpha^{|\tau_x|}|.$$

This distance function can then be extended to all pairs of triangulations by defining $\Delta(\sigma, \tau)$ to be the shortest path distance between σ and τ . One can then define a path coupling and use the coupling to show the following lemma holds.

Lemma 2.1. *There exists $\alpha_0 > 1$ such that the following holds for all $\alpha \geq \alpha_0$, and $\lambda \alpha \leq 1$. Let σ, τ be two triangulations that differ in exactly one edge, and let σ', τ' denote the random triangulation obtained after one step of the Glauber dynamics. Then there exists a coupling of σ' and τ' so that their expected distance satisfies*

$$\mathbb{E}[\Delta(\sigma', \tau')] \leq \Delta(\sigma, \tau) \left(1 - \frac{1}{2|E|}\right),$$

where $|E| = 3mn + m + n$ is the total number of edges in the triangulation.

With this lemma, one can then apply a standard Markov inequality to prove Theorem 2.1. One thing to notice is that the bound in Lemma 2.1 holds only for $\alpha \geq 8$ and $\lambda \leq \alpha^{-1} = 1/8$. Whether a similar upper bound can be proved for $\lambda \in (1/8, 1]$ is still an open problem.

2.5 Some structural properties

In this section, we state without proof some geometric properties of the minimal parallelogram and excluded region that we will use later in the paper. The proofs of these statements can be found in [2].

Proposition 2.2. *Let a, b be positive and coprime, and consider the infinite line through $(0, 0)$ of slope (a, b) . Then the points of $\mathbb{Z}^2$ that are closest to this line (and are not on the line) are at horizontal distance a^{-1} and vertical distance b^{-1} on either side of the line.*

An immediate corollary of the proposition is the following.

Corollary 2.3. *Let a, b be positive and coprime. The infinite family of parallel lines with slope (a, b) , horizontal separation a^{-1} (and thus vertical separation b^{-1}) and such that one line in the family passes through $(0, 0)$ partitions $\mathbb{Z}^2$, in the sense that every integer point lies on one of the lines.*

Now, given any triangulation σ , let σ_x be an arbitrary edge in the triangulation. Without loss of generality, assume σ_x is positively oriented with slope (a, b) where a, b are coprime. Let p_1, p_2 be the two integer vertices of σ_x . Then by Proposition 2.2, there exists a unique closest integer point to the right of σ_x that lies horizontally and vertically between p_1, p_2 . Call this point p_3 . Denote the slope of the line segment $p_3 p_2$ to be (a', b') such that a', b' are coprime. By symmetry, there exists a unique point p_4 on the left of σ_x such that the p_4 is also the unique closest integer point to the left of σ_x that lies between p_1, p_2 horizontally and vertically. Then, p_1, p_2, p_3, p_4 are the four vertices of the minimal parallelogram of σ_x .

Let $\overline{p_i p_j}$ denote the unique straight line that goes through the line segment p_i, p_j . Then, $\overline{p_1 p_3}$ and $\overline{p_2 p_4}$ are parallel to each other and hence, determine a region between them. Call this region $R(\overline{p_1 p_3}, \overline{p_2 p_4})$. Similarly, $\overline{p_1 p_4}$ and $\overline{p_2 p_3}$ also define a region which we denote as $R(\overline{p_1 p_4}, \overline{p_2 p_3})$. Then, the intersection of $R(\overline{p_1 p_3}, \overline{p_2 p_4})$ and $R(\overline{p_1 p_4}, \overline{p_2 p_3})$ is the minimal parallelogram of σ_x . Denote the union of $R(\overline{p_2 p_4}, \overline{p_1 p_3})$ and $R(\overline{p_1 p_4}, \overline{p_2 p_3})$ as the *excluded region* (see Figure 7). Then we have the following result.

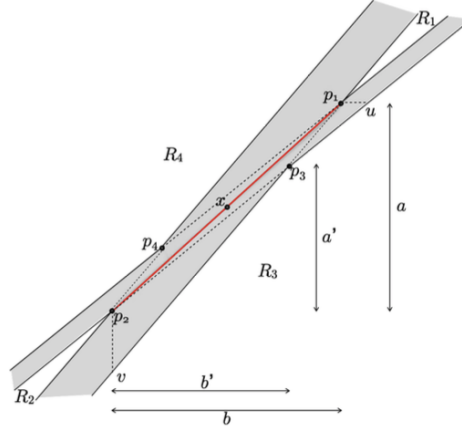


Figure 7: (From [2]) The minimal parallelogram and excluded region of an edge σ_x .

Proposition 2.4. *The excluded region of any nonhorizontal and nonvertical triangulation edge does not contain any integer points in its interior.*

This result is important as it “localizes” the *influence region* of a given edge σ_x . This is particularly useful when we analyze the impact of any long edge in section 6. For the purpose of our discussion in Section 6, we need one last lemma on ground state triangulations with constraints.

Lemma 2.2. (*Ground state lemma*). *Given any set of constraint edges, the ground state triangulation is unique (up to possible flipping of unit diagonals) and can be constructed by placing each edge in its minimal length configuration consistent with the constraints, independent of the other edges.*

3 Lower bounds on the mixing time for $\lambda > 1$

In this section, we reproduce the proof of the current best-known lower bound on the mixing time of the Glauber Dynamics for $\lambda > 1$ by Caputo, Martinelli, Sinclair and Stauffer [2]. Their lower bound on the mixing time exploits the fact that it takes a long time to flip the orientation of a long edge. We say an edge is *positively oriented* (resp., *negatively oriented*) if its leftmost endpoint lies below (resp., above) its rightmost endpoint. A vertical or horizontal edge is neither positively nor negatively oriented.

For $\lambda > 1$, we have the following exponential lower bound on the mixing time of the Glauber dynamics in the absence of constraints.

Theorem 3.1. *For every $\lambda > 1$, there exists $c > 0$ such that the mixing time of the Glauber dynamics on triangulations of $\Lambda_{m,n}^0$ in the absence of constraints satisfies $T_{\text{mix}} \geq \exp(c(m+n))$ for all $m, n \geq 1$.*

The main idea of the proof is to exhibit a set of configurations A that we try to “escape” from. By showing the total weight of the boundary set ∂A of A is small, we can then apply the conductance bound to prove the statement.

Proof. For any $A \subseteq \Omega$ we write

$$Z(A) = \sum_{\sigma \in A} \lambda^{|\sigma|} \quad (1)$$

for the partition function restricted to A . Let ∂A denote the set of $\sigma \in A$ that are adjacent to A^c in the flip graph. Then, the standard conductance bound (see, e.g. [6], Theorem 7.3) will allow us to deduce the theorem if we can find a set A such that $\mu(A) \leq 1/2$ and

$$\frac{Z(\partial A)}{Z(A)} \leq e^{-c(m+n)}.$$

Assume w.l.o.g. that the horizontal coordinate $n \geq m$ (else the same argument applies with the roles of n and m reversed). We define $A \subseteq \Omega$ as the set of triangulations $\sigma \in \Omega$ such that every internal midpoint with half-integer vertical coordinate v is not positively (resp., not negatively) oriented if $v + \frac{1}{2}$ is odd (resp., even).

Moreover, for all $\sigma \in A$, all edges with integer vertical coordinates are horizontal and frozen (unflippable). Notice that any $\sigma \in A$ has the “herringbone” structure as illustrated in Figure 8. In other words, each configuration is divided into 1D “strips” in each horizontal layer. Edges in adjacent layers are oriented in opposite directions (except for possible vertical edges).

The total length of any $\sigma \in A$ can be written as

$$|\sigma| = (m+1)n + (2n+1)m + \sum_{j=1}^m L(\sigma(j)).$$

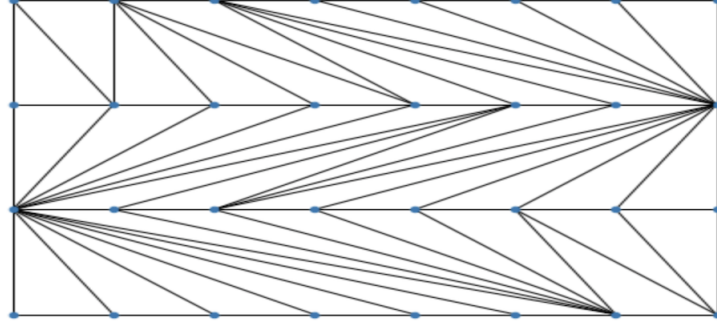


Figure 8: Example of a herringbone-structure triangulation of a 3×7 region

where $(m+1)n$ is the total length of all horizontal (unit length) edges, $(2n+1)m$ is the sum of all the vertical lengths of the edges of σ , and $L(\sigma(j))$ denotes the total horizontal length of the internal edges of the 1D triangulation $\sigma(j)$ in the j th layer.

We let $\Omega_{1,n}^+$ denote the set of all 1D triangulations such that all internal edges are not negatively oriented. Below, we will be using the well-known correspondence between lattice paths and 1D triangulations (see, e.g., [8]). This map is illustrated in Figure 9.

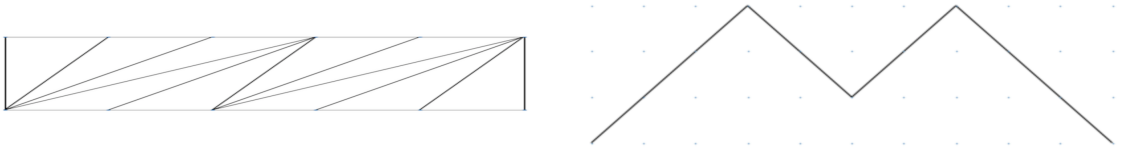


Figure 9: 1D triangulation corresponds to a lattice path

This implies that $\Omega_{1,n}^+$ is in one-to-one correspondence with lattice paths (integer valued sequences with ± 1 increments) of length $2n$, which start at zero, end at zero and stay nonnegative. Moreover, under this correspondence, the total length $L(\pi)$ of $\pi \in \Omega_{1,n}^+$ is precisely the area under the path π . The maximal element in $\Omega_{1,n}^+$, denoted $\hat{\pi}$ has total length

$$L(\hat{\pi}) = \sum_{i=1}^n i + \sum_{i=1}^{n-1} i = n^2.$$

Notice that if $\sigma \in \partial A$, then there exists a layer j such that $\sigma(j)$ contains an internal vertical edge, that is, an edge with horizontal length zero. In particular, on that layer we must have that $L(\sigma(j)) \leq n^2 - n$. Therefore, writing $Z(A), Z(\partial A)$ as products over layers, and summing over the m choices of the distinguished layer j with $L(\sigma(j)) \leq n^2 - n$, one finds

that

$$\begin{aligned} \frac{Z(\partial A)}{Z(A)} &\leq m \frac{\sum_{\pi \in \Omega_{1,n}^+ : L(\pi) \leq n^2 - n} \lambda^{L(\pi)}}{\sum_{\pi' \in \Omega_{1,n}^+} \lambda^{L(\pi')}} \\ &\leq m \sum_{\pi \in \Omega_{1,n}^+ : L(\pi) \leq n^2 - n} \lambda^{L(\pi) - n^2} \leq m \sum_{k \geq n} p_n(k) \lambda^{-k} \end{aligned}$$

where we have estimated $\sum_{\pi' \in \Omega_{1,n}^+} \lambda^{L(\pi')} \geq \lambda^{L(\hat{\pi})} = \lambda^{n^2}$, and we use $p_n(k)$ to denote the number of $\pi \in \Omega_{1,n}^+$ which have $L(\pi) = n^2 - k$. From the correspondence with lattice paths, it is seen that $p_n(k)$ is bounded by the number $p(k)$ of all partitions of the integer k . To see this, observe that

1. If $\pi \in \Omega_{1,n}^+$ with $L(\pi) = n^2 - k$, then the region with area $L(\hat{\pi}) - L(\pi)$ obtained by “subtracting” π from $\hat{\pi}$ is a Young diagram with k boxes.
2. Any Young diagram with k boxes can be seen uniquely as a region obtained by subtraction as above for some (not necessarily nonnegative) lattice path π with area $n^2 - k$.

Since $p(k)$ is the number of Young diagrams with k boxes, restricting to nonnegative lattice paths π one has $p(k) \geq p_n(k)$. Finally, using the well known fact that $p(k) = \exp(O(\sqrt{k}))$, we arrive at the desired bound (1). Now, by the conductance bound, we have thus proved the theorem. $\square$

In the case $m \ll \sqrt{n}$, a slightly stronger lower bound than Theorem 3.1 has been proved. Here we will only state the result without giving a proof. Interested readers can find the proof in [2].

Theorem 3.2. *For every $\lambda > 1$, there exists $c > 0$ such that the mixing time of the Glauber dynamics on triangulations of $\Lambda_{m,n}^0$, in the absence of constraints, satisfies $T_{\text{mix}} \geq \exp(cn^2/m)$ for all $n \geq m^2/c, m \geq 1$.*

It is natural to ask if we can have a better lower bound than the one in Theorem 3.1. Notice that the proof of Theorem 3.1 relies on the one-dimensional structure of the herringbones. This gives rise to the $\exp(\Omega(m+n))$ lower bound in the theorem. Therefore, in an attempt to improve the bound to $\exp(\Omega(mn))$, our intuition tells us to look for patterns that are more “two-dimensional.” This suggests that we should particularly analyze the case when $n \approx m$. We have the following conjecture.

Conjecture 3.3. *For every $\lambda > 1$, there exists $d > 0$ such that the mixing time of the Glauber dynamics on triangulations of $\Lambda_{m,n}^0$ in the absence of constraints satisfies $T_{\text{mix}} \geq \exp(dmn)$ for all $n, m \geq 1$.*

While the analysis in the proof of Theorem 3.1 is tight and hence is not directly applicable to the proof of this conjecture, both the motivations and limitations of “herringbone” configurations provide us with some useful intuitions. The first thing to consider is whether the restricted set A is optimal. Does there exist a larger set B such that $A \subset B$ and it takes $\exp(\Omega(mn))$ to escape the set B ?

Secondly, notice that in the proof of the current lower bound, for simplicity of analysis, each layer of 1D triangulations is oriented in such a way that interactions across multiple layers are not possible unless a given configuration already escapes the restricted set A . It is interesting to ask whether interactions between different layers will weaken or reinforce the restrictive structure.

Lastly, the proof of the current $\exp \Omega(m+n)$ lower bound is built upon a structure that is highly one-dimensional (all interactions between edges occur at some one dimensional “strip”.) This structure is easy to analyze but it does not capture the two-dimensional structure of lattice triangulations. Therefore, in order to obtain an $\exp \Omega(mn)$ lower bound on the mixing time, we need to look for triangulations that exhibit a strong two-dimensional structure.

4 Simulation Methods

In the following sections, we will address the questions raised at the end of Section 3 using results from our simulations. We implemented the algorithms and generated the resulting plots using the Python Programming Language and in particular, the Networkx package.

To cope with the exponential mixing time of the Glauber dynamics, we introduce two speed-up simulation methods both of which maintain the same expected number of attempted steps as the Glauber dynamics in the actual simulation. If we have a speed-up algorithm which selects an edge to flip uniformly at random and flip it deterministically, then it only runs one actual step (pick an edge and flip it) but the effect of this move (an edge is flipped) is only realized in, for instance, 10 steps in expectation in the original Glauber dynamics. In this case, we say that the algorithm uses 1 *actual step* to simulate 10 *attempted steps*. Let $T(C)$ be the random variable which denotes the number of *attempted steps* for the Glauber dynamics to flip one edge of the configuration C and $t(C)$ be the number of *actual steps* in the simulation. Let $T_i(C)$ and $t_i(C)$ denote the same quantities under speed-up methods $i = 1, 2$. Then, $\mathbb{E}[T_i(C)] = \mathbb{E}[T(C)]$ for $i = 1, 2$ and $t_i(C) = \text{poly}(n)^{-1} t(C)$ where n is the dimension of the triangulation graph. Crucially, both speed-up methods respect the same conditional probabilities of the edge being flipped as the original Glauber dynamics. We now give a brief description and analysis of the methods given a configuration C .

1. (The Glauber dynamics) In the original Glauber Dynamics, we select an edge (mid-point) uniformly at random and flip it with its corresponding probability. Let N denote the total number edges in the graph and n_f denote the number of flippable edges. Denote the flippable edges by $e_1, \dots, e_{n_f}$ and the corresponding flip probabilities by $p_1, \dots, p_{n_f}$. Then the random variable $T(C)$ follows the geometric distribution with parameter $\sum_{i=1}^{n_f} p_i/N$. Then it follows that $\mathbb{E}[T] = N / \sum_{i=1}^{n_f} p_i$ and $\text{var}(T) =$

$\frac{1 - \frac{\sum p_i}{N}}{(\sum p_i/N)^2}$ and the probability of edge e_i being flipped given a flip has occurred is $\mathbb{P}(e_i \text{ flipped} \mid \text{A flip has occurred}) = \frac{p_i}{\sum_{j=1}^{n_f} p_j}$.

2. (Alternative Method 1) Instead of selecting an edge which may not be flippable, we select an edge from the set of flippable edges uniformly at random and flip it with its corresponding flip probability. Following the same notation as in the analysis above, the probability that edge e_i is flipped is p_i/n_f . Hence, the probability that any edge is flipped is $\sum_{i=1}^{n_f} p_i/n_f$. Since we are selecting from the flippable edges only, we are omitting the number of attempted steps needed to select a flippable edge in the original Glauber dynamics. To compensate for this, each time we select a flippable edge, we add N/n_f to the number of attempted steps. In other words,

$$T_1(C) \sim \frac{N}{n_f} \text{Geom} \left(\frac{\sum_{i=1}^{n_f} p_i}{n_f} \right).$$

Hence, $\mathbb{E}[T_1(C)] = \frac{N}{n_f} \cdot \frac{n_f}{\sum p_i} = \frac{N}{\sum p_i} = \mathbb{E}[T(C)]$. Moreover, $\mathbb{P}(e_i \text{ flipped} \mid \text{a flip has occurred}) =$

$\frac{\frac{1}{n_f} p_i}{\frac{1}{n} \sum p_j} = \frac{p_i}{\sum p_j}$. Let G be a random variable such that $G \sim \text{Geom} \left(\frac{\sum_{i=1}^{n_f} p_i}{n_f} \right)$. Then, the

variance is given by:

$$\text{var}(T_1(C)) = \frac{N^2}{n_f^2} \text{var}(G) = \frac{N^2}{n_f^2} \frac{1 - \frac{\sum p_i}{n_f}}{(\sum p_i/n_f)^2} = \frac{1 - \frac{\sum p_i}{n_f}}{(\sum p_i/N)^2}$$

Since $n_f < N$, $\text{var}(T_1(C))$ is strictly less than $\text{var}(T(C))$. The speed-up achieved by this method is

$$t_1(C) = \frac{n_f}{N} t(C)$$

3. (Alternative Method 2) To further speed up the simulation, we randomly select a flippable edge and flip it with probability one. To retain the same conditional distribution, the probability of a given flippable edge e_i being selected is $p_i/\sum p_j$. Similar to alternative method 1, we compensate for the omitted attempted steps in the original Glauber dynamics by adding $\frac{N}{\sum p_i}$ to the total number of attempted steps. Notice that $T_2(C)$ is a constant in this method and hence $\mathbb{E}[T_2(C)] = \frac{N}{\sum p_i} = \mathbb{E}[T(C)]$ and $\text{var}(T_2(C)) = 0$. The speed-up achieved by this method is given by:

$$t_2(C) = \frac{\sum p_i}{N} t(C) < \frac{n_f}{N} t(C)$$

Both of the alternative methods achieve a polynomial factor speed-up while maintaining the right expectation and conditional probabilities. Consequently, both can be used to simulate

the long-term behavior of the Glauber dynamics. Notice that Alternative Method 2 is faster than Alternative Method 1 as at each attempted step it is guaranteed to “make progress” since it is deterministic. The drawback to Alternative Method 2 is that the number of attempted steps before an effective step is a constant. This is not ideal since the original Glauber Dynamics has a variance of $\text{var}(T) = \frac{1 - \frac{\sum p_i}{N}}{(\sum p_i/N)^2}$. Therefore, Alternative Method 2 is not appropriate if we want to quantitatively compare the number of attempted steps before a certain event happens.

For tasks in the simulations below that involve quantitative analysis of the number of attempted steps, we will use Alternative Method 1. On the other hand, for tasks that involve only the long-term behavior of the Dynamics, we will use Alternative Method 2 for the magnitude of speed-up it provides.

We will use a few metrics like average edge length, total flippability and orientations to compare different configurations.

5 Simulation studies of the herringbone and the uni-directional herringbone structures

5.1 The uni-directional herringbone structure

Motivated by the questions raised at the end of Section 3, we will perform simulations on both the triangulations with the herringbone structure and those with the *uni-directional* herringbone which we will define shortly. In particular, we want to study and compare the long-term behaviors of the herringbone and the uni-directional herringbone triangulations.

In the definition of the herringbone structure, we specify that edges in adjacent layers are oriented in different directions. We define the *uni-directional herringbone structure* similarly except we let edges in all layers to be oriented in the same direction. Specifically, we define a triangulation σ with the positively-oriented (resp., negatively oriented) *uni-directional herringbone* structure to be triangulations such that for all edges with integer vertical coordinate are horizontal unit edges and every internal midpoint with half-integer vertical coordinate is not negatively oriented (resp., not positively oriented.) See Figure 10 for an example.

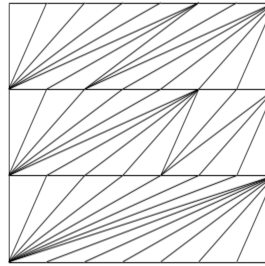
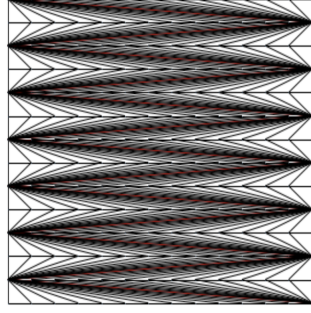
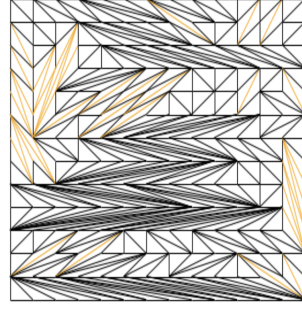


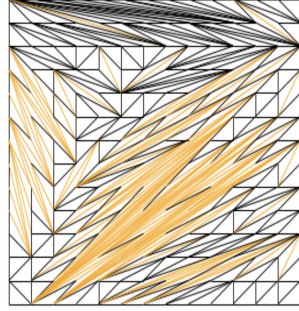
Figure 10: Example of a positively-oriented uni-directional herringbone structure triangulation of a 3×7 region



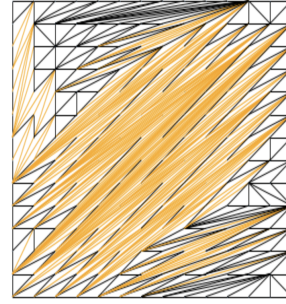
(a) State 1 (Initial configuration): A 13×13 triangulation with the herringbone structure



(b) State 2: The herringbone structure breaks in some layers; edges with vertical length > 2 begin to form



(c) State 3: Long diagonal edges begin to form



(d) State 4: More long diagonal edges are formed

Figure 11: Simulation of a 13×13 herringbone configuration

5.2 Simulations

5.2.1 Simulation on herringbones

We first use Alternative Method 2 to study the long-term behaviors of the Glauber dynamics on triangulations with the herringbone structure. We then use the same method to study the long-term behavior of triangulations with the uni-directional herringbone structure. The reason we are using Alternative Method 2 is because we are interested in the long-term behavior of the Glauber dynamics and Alternative Method 2 allows us to simulate long-term behaviors in a relatively short amount of time. However, as mentioned in Section 4, Alternative Method 2 cannot capture the variance of the number attempted steps in the original Glauber dynamics. Therefore, we will not draw any conclusions on the number of attempted steps. We will only be describing the long-term behavior of the dynamics.

We first simulate a 13×13 triangulation with the herringbone structure. The main purpose of this simulation is to observe the behaviors of the configuration after it escapes

from the boundary set defined in Section 4. We include a sequence of figures that are representative of the behavior of the dynamics in the long run in Figure 11 where the edges that have vertical length ≥ 2 are colored in yellow. For the remainder of this section, we will call these edges “2d-edges.”

From state 1 (the initial configuration) to state 2, we observe that the herringbone structure is breaking down as we see several layers containing the ground state unit squares. We also see that 2d-edges begin to form in layers where the herringbone structure breaks. Notice that for layers that haven’t completely broken down yet, they seem to maintain the one-dimensional herringbone structure in the middle but not on the two ends. In other words, triangulations with the herringbone structure tend to break from the two ends instead of breaking from the middle.

From state 2 to state 3, there are a couple of things worth mentioning. First, we see that the herringbone structure is occurring again in regions where the herringbone structure has already broken down (the top rows of the configuration). This is reasonable since flips are reversible and long edges are favored in the scenario where $\lambda > 1$. Second, a somewhat surprising phenomenon is the emergence of long diagonal edges oriented in the same direction. This tendency is further reinforced from state 3 to state 4. This seems to suggest that eventually the herringbone structure is “absorbed” in the configuration with long diagonal edges oriented in the same direction.

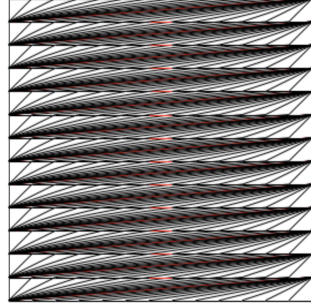
It seems from this simulation that the herringbone structure undergoes two phases of transitions in the long run: it first breaks its one-dimensional structure and then edges begin to aggregate and form long diagonal edges in the same direction. Because of this two-phase phenomenon, we cannot deduce from these simulations that Phase 1 and Phase 4 are contained in the same set of larger configurations A that is hard to escape from. But still, these simulations give us some intuitions about potential candidates for an absorbing set of configurations that is hard to escape from. Such an absorbing set seems to include configurations that contain many long diagonal edges oriented in the same direction.

5.2.2 Simulations on uni-directional herringbones

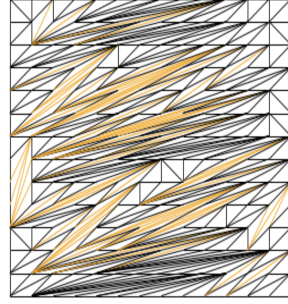
Now we switch our attention to herringbones with the uni-directional structure. Similar as above, we will simulate a configuration with the uni-directional herringbone structure for a long time using Alternative Method 2. We will observe the long-term behaviors of this particular structure and compare them with the long-term behaviors of the herringbone structure. See Figure 12 for the simulations results.

From state 1 to state 2, we see that “2d-edges” begin to form. However, unlike in the herringbone case, the one-dimensional structure breaks both at the two ends and in the middle. We see “2d-edges” emerge in the middle of two layers such that the rest of the edges in the two layers maintain the same one-dimensional structure as in the herringbone structure (see the top two layers in State 2.)

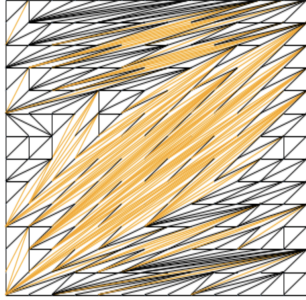
From state 2 to state 3 and eventually to state 4, we observe that the transitions seem to be “monotone.” The common tendency in these states is that in the long run, long edges oriented in the same direction tend to form regions of long diagonal edges. These edges are roughly $2n$ in length which is the longest an edge can be in any $n \times n$ triangulation. Notice that in this sequence of transitions, some small regions of unit squares and short



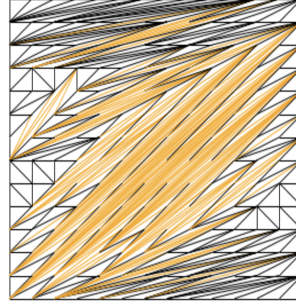
(a) State 1 (Initial configuration): A 13×13 triangulation with the uni-directional herringbone structure



(b) State 2: The herringbone structure breaks in some layers; edges with vertical length > 2 begin to form



(c) State 3: Long diagonal edges begin to form



(d) State 4 (End configuration): More long diagonal edges are formed

Figure 12: Simulation of a 13×13 positively-oriented uni-directional herringbone configuration

edges oriented in the opposite direction can occur.

The monotone behaviors of the uni-directional herringbone structure in the long run suggest that while the one-dimensional structure seems to break, the resulting configurations are absorbed into configurations with long diagonal edges oriented in the same direction. Since the uni-directional herringbone structure we started with is also comprised of edges oriented in the same direction, we conjecture that both the initial configuration and the end configuration in Figure 12 may belong to the same set of configurations B . The defining characteristics of B seems to be long diagonal edges oriented in the same direction. However, the actual definition of the set B seems to be rather subtle as it also has to tolerate some regions of unit squares and short edges oriented in the opposite direction.

6 Simulation studies of fan configurations

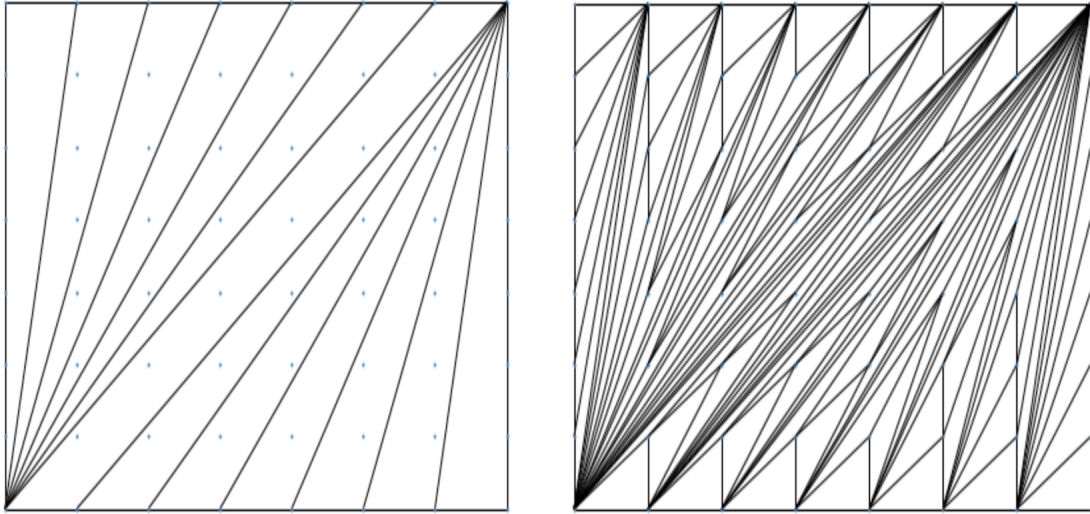
6.1 The fan configuration

Motivated by our observations towards the end of Section 5, we now define a set of configurations with long edges oriented in the same direction. We then perform some simulations to show that experimentally it takes a long time to escape from these configurations.

Given an $m \times n$ rectangular region of the integer lattice, we denote each vertex with its corresponding coordinates. For instance, the lower-left vertex is denoted as $(0,0)$ and the upper-right vertex is denoted as (n,m) . Let (i,j) denote the greatest common divisor of i,j . We then define a positively oriented *fan triangulation* σ as follows:

1. For each $i \in \{1, \dots, n\}$ such that $(i, n) = 1$, add the edge $((0,0), (i,n))$ to σ .
2. For each $j \in \{1, \dots, n\}$ such that $(j, n) = 1$, add the edge $((j,0), (n,m))$ to σ .
3. Treating the edges included in step 1 and step 2 as constraints, add edges that are not negatively oriented and consistent with the constraints to σ until a full triangulation is achieved.

We can similarly define a negatively oriented fan configuration. The intuition behind this construction is to include as many long edges in the same direction as possible. The condition $(i,n) = (j,n) = 1$ makes sure that the edges we add in step 1 and 2 do not go through any other half lattice points except for their endpoints.



(a) After step 1 & 2

(b) After step 3

Figure 13: A 7×7 fan configuration

Notice that we can always add non-negatively oriented edges only for positively oriented fan configurations (resp. non-positively oriented edges only for negatively oriented fan configurations) to complete the full triangulation in step 3. To see this, by the ground

state lemma (Lemma 2.2), there exists a unique ground state configuration consistent with the constraints (up to possible flipping of unit diagonals). We can orient all unit diagonals in the positive direction (resp. negative direction) and then the rest of the edges have to be in the same direction as the constraints since one has to go through a sequence of flips that include the unit diagonals to the opposite direction. See Figure 13 for an example of a fan configuration.

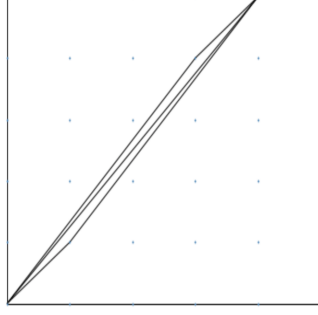
We are interested in the case when $n \approx m$ for reasons mentioned in Section 3. Let's assume that $m = n$. Notice that with the edges added in step 1 and step 2, we have at least $2\phi(n)$ edges of length at least $n + 1$ where $\phi(n)$ is Euler's Totient function which denotes the number positive integers smaller than n that is coprime to n . Use $\{\sigma_{x_i}\}_{i=1}^{2\phi(n)}$ to denote these long edges. Then σ_{x_i} is flippable to a shorter edge for all $i = 1, \dots, 2\phi(n)$. To see this, we know that each edge is of length greater than n . Therefore, by Proposition 2.2 and its corollary, there exists one and only one vertex on each side of e_i of vertical coordinate between 0 and n that is closest to e_i . The two endpoints of σ_{x_i} together with these two closest points on each side of σ_{x_i} form the minimal parallelogram of σ_{x_i} . As a result, σ_{x_i} is flippable to a shorter edge for all $i = 1, \dots, 2\phi(n)$.

Although each of these long edges σ_{x_i} is flippable to a shorter edge σ'_{x_i} , the length decrease varies from 2 to $O(n)$. We state this result in the following proposition.

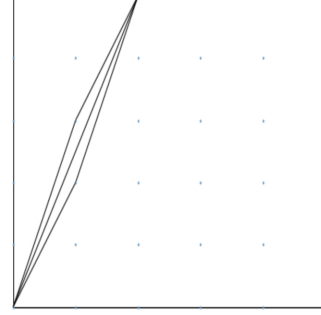
Proposition 6.1. *Let σ_{x_i} be a long edge defined in step 1. Then σ_{x_i} is flippable and it can be flipped to a shorter edge σ'_{x_i} . Moreover, the decrease in length $|\sigma'_{x_i} - \sigma_{x_i}|$ varies in the range $[2, n - 2]$ depending on the slope of σ_{x_i} .*

Proof. For a given long edge σ_{x_i} , let its slope be n/k such that $k \in \{1, \dots, n-1\}$ and $(n, k) = 1$. Then by Proposition 2.2, the points closest to this line are of vertical distance $1/k$ and horizontal distance $1/n$. Let $(a, b) \in \mathbb{Z}^2$ be such a point to the right of σ_{x_i} , then $(a, b + 1/k) \in L$ and $(a - \frac{1}{n}, b) \in L$ where $L = \{(x, y) \in \mathbb{R}^2 | y = \frac{n}{k}x\}$. Then $\frac{n}{k}a = b + 1/k$ and $\frac{n}{k}(a - 1/n) = b$. Therefore, one must have that $b = \frac{na-1}{k}$. Since $b \in \mathbb{Z}$, it must hold that $na - 1 \equiv 0 \pmod{k}$. Because $\gcd(n, k) = 1$, there exists a unique inverse of n in $\mathbb{Z}_k$. As a result, depending on the value of k , the value $n^{-1} \pmod{k}$ varies and hence the decrease in length $|\sigma'_{x_i} - \sigma_{x_i}|$ varies as well. See Figure 14. □

This result can be easily extended to the case when $m \neq n$. For the subsequent simulations, we will assume for simplicity that $n = p$ for some prime number p . In this case, since $\phi(p) = p - 1$, all possible long edges are included. Notice that these $2(p - 1)$ long edges partition the $n \times n$ square region into $2(p - 1)$ triangular regions of area $n/2$ and one parallelogram of area n along the diagonal. We will first perform simulation studies on the long triangulations for $n = p$ for some prime p , and then we will perform similar simulations for the case when n is not prime to argue the result is robust.



(a) Example of σ_x such that $|\sigma'_{x_i} - \sigma_{x_i}|$ is small ($= 6$)



(b) Example of σ_x such that $|\sigma'_{x_i} - \sigma_{x_i}|$ is large ($= 2$)

Figure 14: Long edges in a 5×5 square region

6.2 Simulations

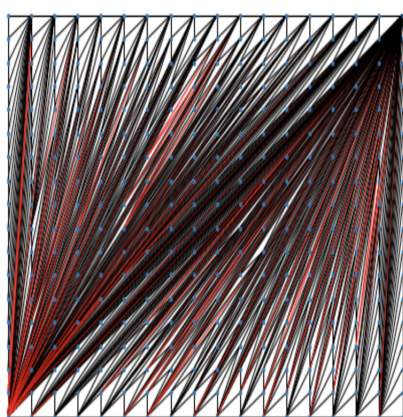
We use Alternative Method 1 to simulate the behavior of the Glauber dynamics since we want to analyze the behavior of the dynamics after a certain number of attempted steps. There is a tradeoff we can make: we can either simulate a large configuration (n big) for a small number of attempted steps or we can simulate a smaller configuration (n small) for a larger number of steps. The advantage of the former simulation is that it may reveal some structures of the configuration that are not present in a smaller size configuration. The advantage of the latter is that it allows us to view the long-term behavior as it can simulate a lot more steps. The strategy we will follow is to first simulate the dynamics on a larger configuration for structural behaviors and then simulate the dynamics on a smaller configuration for the long-term behavior.

6.2.1 Simulation on a 17×17 fan configuration

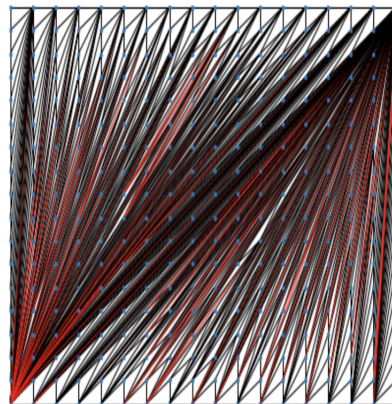
We first simulate the dynamics with the initial triangulation being a 17×17 long triangulation as in Figure 15. The red edges are flippable edges and the black edges are non-flippable edges. This initial configuration has the following statistics: all of its edges are oriented non-negatively as desired; there are a total of 67 flippable edges, 64 of which are flippable to shorter edges and 3 of which are flippable to longer edges; the number of edges flippable to longer versus the number of edges flippable to shorter ratio is 0.047; the average length of all edges is 8.735. See Figure 15.

After 1,000,000 attempted steps. The resulting configuration has the following statistics: all of its edges are still oriented non-negatively; there are a total of 66 flippable edges, 65 of which are flippable to shorter edges and 1 of which are flippable to longer edges; the number of edges flippable to longer versus the number of edges flippable to shorter ratio is 0.015; the average length of all edges is 8.801. The resulting configuration is in Figure 15.

The above simulation results indicate that after 1,000,000 attempted steps, the resulting configuration seems to reinforce the structure of the initial configuration as it has a larger proportion of edges that are flippable to shorter edges than that in the initial configuration. The average length becomes slightly larger as well. Moreover, as we can see in Figure 15,



(a) Initial configuration: a long 17×17 triangulation



(b) End configuration: a long 17×17 triangulation after 1,000,000 steps of the Glauber dynamics

Figure 15: Simulation of a 17×17 fan configuration

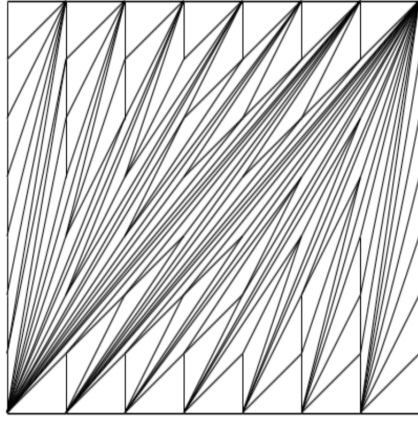
long edges are preserved. This does not tell us much about whether the mixing time of the dynamics starting at a fan configuration is $\exp(\Omega(n^2))$ since the number of attempted steps 1,000,000 is not greater than $\exp(17) \approx 20,000,000$. But our simulation on a 7×7 fan configuration seems to indicate a mixing time of $\exp(\Omega(n^2))$.

6.2.2 Simulation on a 7×7 fan configuration

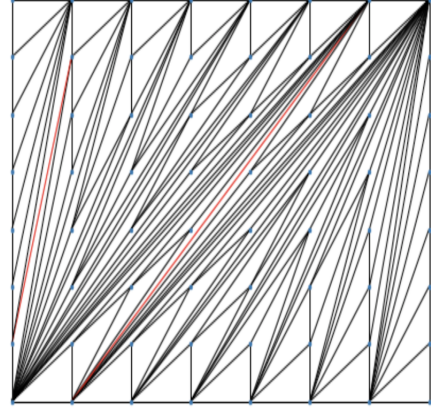
Now we simulate the dynamics with the initial triangulation being a 7×7 long triangulation as in Figure 16. This initial configuration has the following statistics: all of its edges are oriented non-negatively as desired; there are a total of 13 flippable edges; all of them are flippable to shorter edges; the average length of all edges is 4.050. See Figure 16.

After 100,000,000 attempted steps, the resulting configuration has the following statistics: all of its edges are still oriented non-negatively; there are a total of 15 flippable edges, 13 of which are flippable to shorter edges and 2 of which are flippable to longer edges; the number of edges flippable to longer versus the number of edges flippable to shorter ratio is 0.154; the average length of all edges is 4.025. The resulting configuration is in Figure 16.

Based on the pictures and the statistics, the resulting configuration is almost identical to the initial configuration. In other words, the configuration does not change much after 100,000,000 attempted steps. This simulation experimentally adds credibility to our hypothesis about the mixing time of the Glauber dynamics on triangulations of $\Lambda_{m,n}^0$ is potentially larger than a $\exp(\Omega(n))$ in the absence of constraints.



(a) Initial configuration: a long 7×7 triangulation



(b) End configuration: a long 7×7 triangulation after 100,000,000 steps of the Glauber dynamics

Figure 16: Simulation of a 7×7 fan configuration; edges in red are those that are different in the end configuration than in initial configuration

6.2.3 Simulation on a fan configuration when n is not prime

In this section, we perform simulations on the fan configurations when n is not a prime. The goal is to show that the simplifying assumption $n = p$ for some prime p does not affect the overall structure of the configuration too much. In order to achieve this goal, we perform simulations on an 8×8 fan configuration and simulate the same number of attempted steps as we did for the 7×7 fan configuration. We will then compare the behaviors of the dynamics on these two fan configurations corresponding to n being a prime and not a prime.

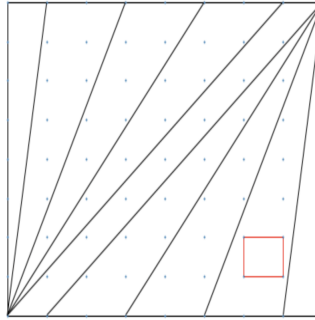
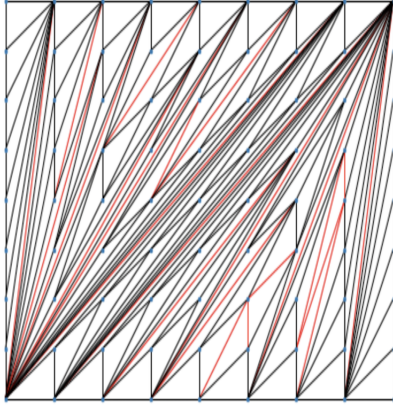


Figure 17: Long edges added in step 1 for an 8×8 long triangulation (the square in red is a unit square that is allowed between two long edges)

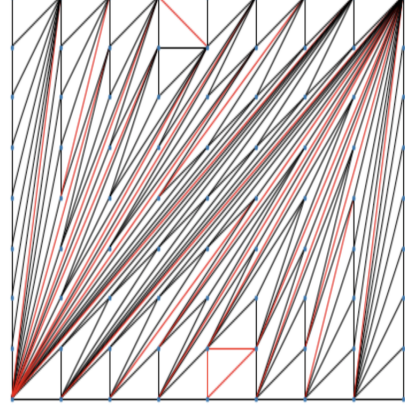
One main structural difference when n is not a prime is that it allows wider triangular regions between the long edges. Notice that with the larger “free” regions between long edges, it is possible for edges inside these “free” regions to be oriented in the opposite direction even though the long edges have not yet been flipped. See, for instance, Figure 17.

Now we perform simulations on the 8×8 fan configuration. The initial configuration (see Figure 18) has the following statistics: all edges are oriented in the same direction; there are a total of 23 flippable edges and the average edge length is 4.038.

After 100,000,000 attempted steps, the resulting configuration has the following properties: there is one edge oriented in the opposite direction; the total number of flippable edges is 24 and the average edge length is 4.308. See Figure 18.



(a) Initial configuration: a long 7×7 triangulation



(b) End configuration: a long 8×8 triangulation after 100,000,000 steps of the Glauber dynamics

Figure 18: Simulation of a 8×8 fan configuration

6.3 Implications of the simulation results

The result shows that a fan configuration when n is not prime exhibits the same behavior as a fan configuration when n is prime. One important implication of this is that when one is trying to find a set of configurations that is hard to escape from, one can allow some small regions that contain edges oriented in the opposite direction. This is also supported by the simulation results in Section 5. One possible way to improve upon the current best lower bound is to define a less restrictive set A which is defined in the proof of Theorem 3.1. The simulation results suggest that we should define a set A that has the following properties:

1. The set A should contain configurations that have at least a certain number of long edges oriented in the same direction.
2. Configurations that contain only a small number of short edges oriented in the opposite direction should be included in A .
3. A configuration should not belong in A if it has a long edge oriented in the opposite direction.

Notice that both the uni-directional 1D configurations and the fan configurations we defined earlier satisfy these properties. However, configurations with a herringbone structure does not belong to this category since they contain long edges in both directions.

The properties we mentioned above give us some intuition about possible approaches to improving the lower bound. But there still exist major difficulties analyzing the configurations. The first difficulty is that the properties listed above are hard to specify exactly. Although we know intuitively that long edges in the same orientation should be included in this less restrictive set A , we do not know exactly how long these edges should be. Similarly, it is difficult to define the exact number and length of short edges we can tolerate in the opposite direction.

The other difficulty is the non-local and non-monotone impact of the flips. To see this, let's look at Figure 19. In this figure, we have five flippable edges in total which we call L_1, C, L_2, R_2, R_1 colored in red, blue, green, orange, purple respectively. All of the edges in the figure are oriented in the positive direction and are valid as edges in a triangulation. Let $\Delta(\sigma_x) = |\sigma'_x| - |\sigma_x|$ be the change in length of the triangulation after σ_x is flipped. To examine the impact of the flips, see Table 1.

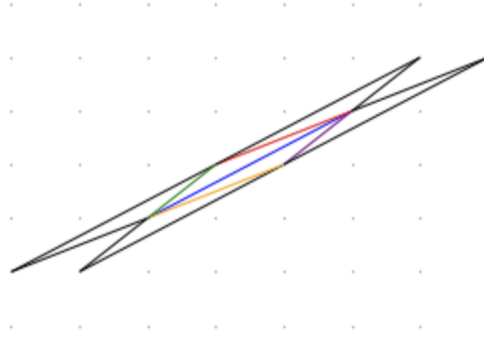


Figure 19: Example of the non-local and non-monotone impacts of a flip

We want to see whether a given flip brings us closer to any ground state configuration so as to escape from the set of configurations with long diagonal edges oriented in the same direction. This is a natural thing to check since ground state configurations do not contain long diagonal edges. From Table 1, we notice that although edge C can be flipped to a shorter edge which brings us closer in length to a ground state triangulation, we lose a lot of “flippability” if we flip C since locally the only edge that remains flippable is C itself. This reduces the number of possible locations of future flips, which brings us farther from any ground state configuration. On the other hand, edge L_1, L_2, R_2 and R_1 are all flippable

Edge flipped	Current length	New length	$\Delta(\sigma_x)$	Edges that remain flippable
L_1 (Red)	3	7	4	L_1, R_1, R_2
C (Blue)	5	1	-4	C
L_2 (Green)	2	8	6	$L_2 R_1 R_2$
R_2 (Orange)	3	7	4	$R_2 L_1 L_2$
R_1 (Purple)	2	8	6	$R_1 L_1 L_2$

Table 1: Summaries of the impacts of flips in Figure 19

to longer edges. This is not desired since ground state triangulations have small average edge length. However, we still have three edges that are flippable after the flip which is more favorable than having fewer flippable edges.

Therefore, we have seen that edge flips are not local in that they not only affect the edge that is being flipped, but also affect the flippability of surrounding edges. Moreover, we have also seen that the impact of an edge flip is not obviously monotone since it brings the triangulation both closer to and farther away from any ground state configuration depending on what the notion of “distance” is.

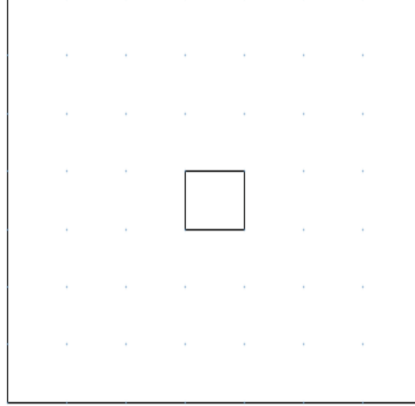
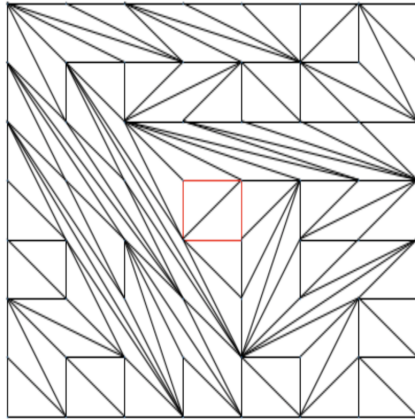
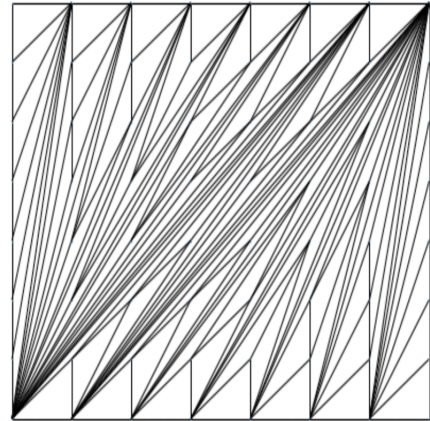


Figure 20: Example of a square constraints in the middle of a 7×7 square



(a) A triangulation of a 7×7 region with square constraints in the middle (in red). The average edge length of this triangulation is 2.11.



(b) A triangulation of a 7×7 region with the fan configuration. The average edge length of this triangulation is 4.05.

Figure 21: Triangulations of a 7×7 square region with and without square constraints in the middle

Another key ingredient in the proof of Theorem 3.1 we can possibly exploit is that the

presence of a vertical edge somewhere in an one-dimensional strip reduces the maximum length of triangulations in that strip by at least n . This is crucial in the proof since the boundary set ∂A is defined to be exactly the set of triangulations with the herringbone structure that have at least one vertical edge. Following the same logic, if we can show that by placing a square somewhere in the middle of the rectangle whose edges serve as constraints (see Figure 20), the maximum total length (or equivalantly, the average length) of the configuration would decrease by $\Omega(n^2)$, then we could possibly follow the proof of Theorem 3.1 with an appropriate definition of the set A and ∂A . We illustrate this point in Figure 21.

7 Conclusion

Through our simulation studies, we examined characteristics of certain configurations that are useful for proving the lower bounds on the mixing time of the Glauber dynamics when $\lambda > 1$. In particular, we conjectured that configurations which contain long diagonal edges in the same direction may be in a large set of configurations that is hard to escape from. However, the difficulty is to give a proper definition of this set. Future work is needed to theoretically confirm the observations from the simulation results.

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